

CAP 4. CAMADA DE APLICAÇÃO

AULA 13: FTP

INE5422 REDES DE COMPUTADORES II

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[HTTPS://MOODLE.UFSC.BR](https://moodle.ufsc.br)

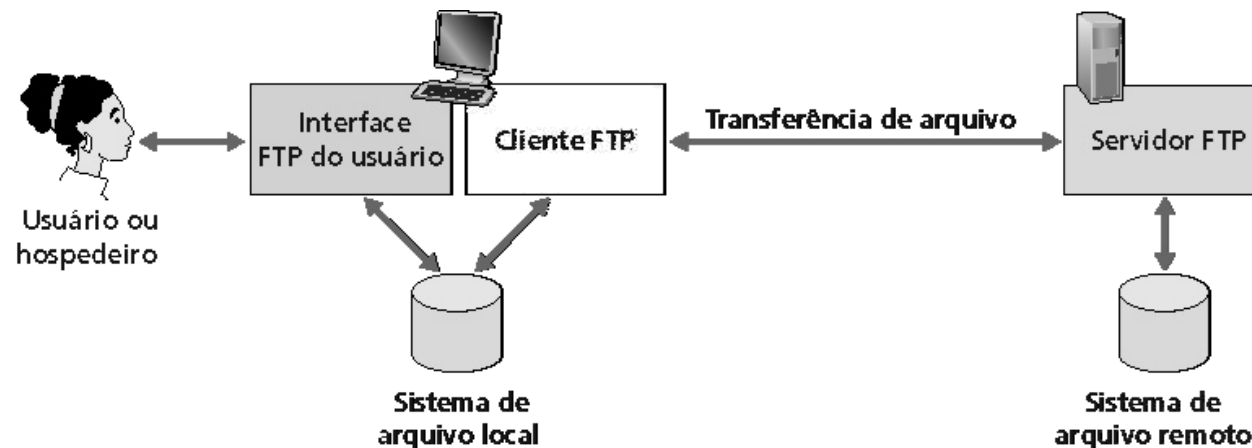
FTP: File Transfer Protocol

FTP (RFC 959) permite

- Transferir, renomear ou remover arquivos remotos
- Criar, remover e modificar diretórios remotos

Modelo cliente/servidor

- Cliente inicia transferência passando o nome (login) e sua senha
- Servidor: host remoto
 - Acessível via a porta 21
 - Servidor usa porta 20 para conexão de dados (modo ativo)



FTP: conexões separadas p/ controle, dados

Conexão de Controle (Porta 21)

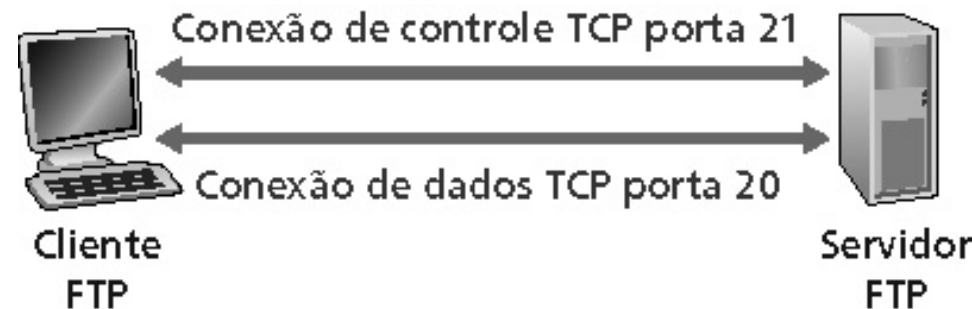
- Aberta pelo cliente FTP
- Cliente obtém autorização através da conexão de controle
- Cliente consulta o diretório remoto enviando comandos através da conexão de controle

Conexão de dados (porta 20)

- Servidor (modo ativo) ou Cliente (modo passivo) abre uma segunda conexão TCP para transferir dados
- Após a transmissão de um arquivo o servidor fecha a conexão de dados

Servidor FTP mantém o “estado”:

- diretório atual, autenticação anterior



FTP: File Transfer Protocol

Duas formas de se conectar a um servidor FTP

- Forma autenticada (nome do usuário e password)
 - direitos de acesso do usuário
- Forma anônima
 - Nome de login: anonymous
 - Password: E-mail
 - Têm-se acesso a repositórios públicos de arquivos
 - qualquer pessoa pode acessar
 - não é preciso cadastrar-se
 - Direitos autorais:
 - repositórios públicos contém apenas arquivos em domínio público

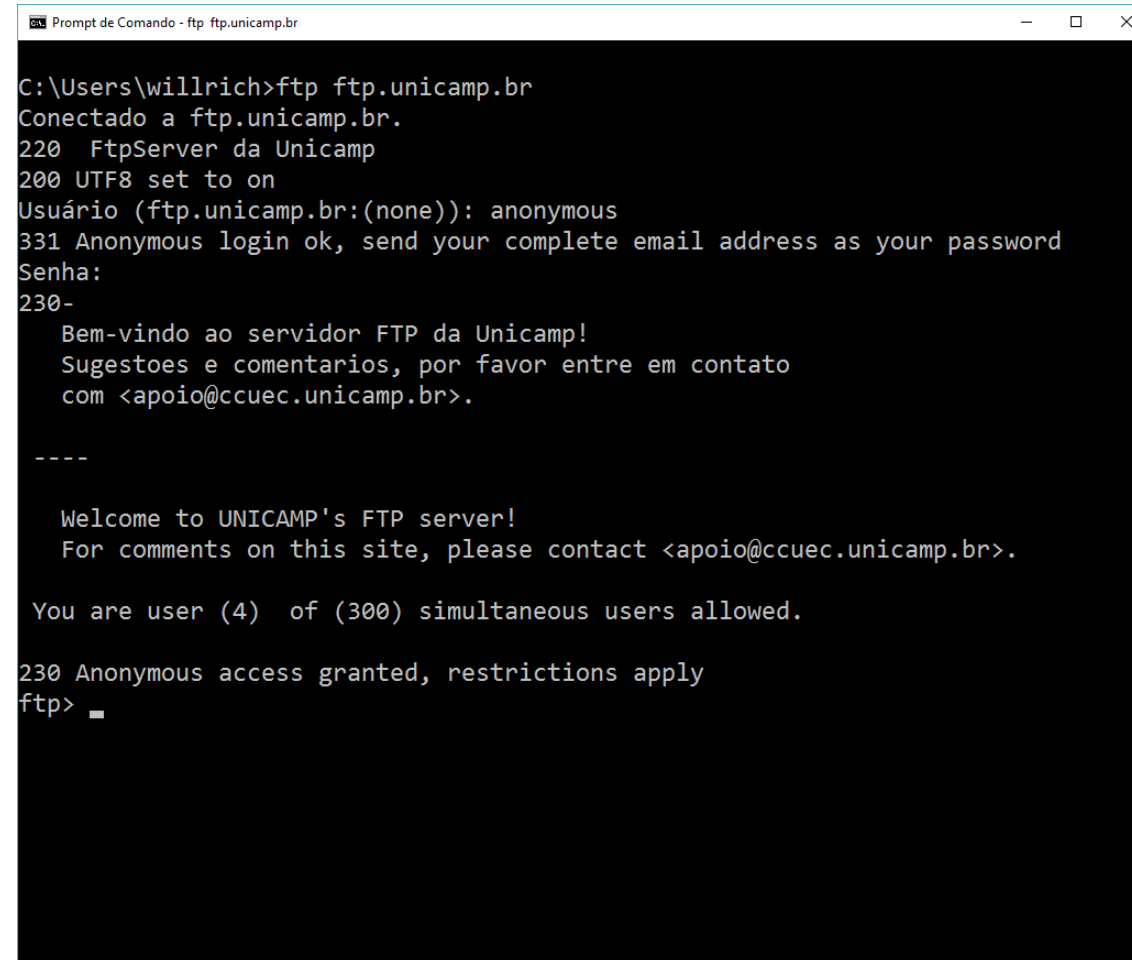
FTP: File Transfer Protocol

Programa FTP

- Implementa o protocolo FTP
- Existem várias implementações fornecendo interfaces gráficas ou não

Forma não gráfica (console)

- Conectando a uma máquina: ftp nome-da-máquina-remota



```
Prompt de Comando - ftp ftp.unicamp.br

C:\Users\willrich>ftp ftp.unicamp.br
Conectado a ftp.unicamp.br.
220 FtpServer da Unicamp
200 UTF8 set to on
Usuário (ftp.unicamp.br:(none)): anonymous
331 Anonymous login ok, send your complete email address as your password
Senha:
230-
    Bem-vindo ao servidor FTP da Unicamp!
    Sugestoes e comentarios, por favor entre em contato
    com <apoio@ccuec.unicamp.br>.

----

    Welcome to UNICAMP's FTP server!
    For comments on this site, please contact <apoio@ccuec.unicamp.br>.

    You are user (4) of (300) simultaneous users allowed.

230 Anonymous access granted, restrictions apply
ftp> _
```

FTP: File Transfer Protocol

Comandos FTP

- Conhecendo o conteúdo das máquinas via ftp
 - ftp> dir [nome-do-diretório] [nome-do-arquivo]
 - ftp> ls [nome-do-diretório] [nome-do-arquivo]
 - ftp> !dir (LOCAL)
- Trabalhando com diretórios
 - ftp> lcd [nome-do-diretório] (Local)
 - ftp> cd [nome-do diretório] (Remoto)
 - ftp> pwd (Diretório remoto atual)
/home/venus/willrich

FTP: File Transfer Protocol

Comandos FTP

- Transferência de arquivos de texto (ASCII) e binários

```
ftp> binary
```

```
200 Type set to I.
```

```
ftp> ascii
```

```
200 Type set to A.
```

- Exemplos de arquivos e modos de transferência
 - Binários
 - Programas, etc...
 - ASCII
 - texto, mensagens de correio eletrônico, PostScript, etc...

FTP: File Transfer Protocol

Comandos FTP

- Transferindo arquivos: comandos get e put

```
ftp> get arquivo-fonte [arquivo-destino]
```

```
ftp> put arquivo-fonte [arquivo-destino]
```

- Exemplo de transferência de arquivo:

```
ftp> get comentario
```

```
200 PORT command successful.
```

```
150 ASCII data connection for comentario (150.162.60.1,3516) (1588 bytes)
```

```
226 ASCII Transfer complete.
```

```
1634 bytes received in 0.052 seconds (30 Kbytes/s)
```

```
ftp> quit
```

```
221 Goodbye.
```


Protocolo FTP: comandos, respostas

Comandos típicos:

- enviados em texto ASCII pelo canal de controle
- USER nome
- PASS senha
- LIST devolve lista de arquivos no diretório atual
- RETR arquivo recupera (lê) arquivo remoto
- STOR arquivo armazena (escreve) arquivo no hospedeiro remoto
- PASV indica o modo passivo
- PORT h1, h2, h3, h4, p1, p2 porta informa a porta de dados
 - H1.h2.h3.h4 é o endereço IP
 - P1*256+p2 é a porta

Códigos de retorno típicos

- código e frase de status (como para http)
- 331 Username OK, password required
- 125 data connection already open; transfer starting
- 425 Can't open data connection
- 452 Error writing file

Tente: telnet ftp.unicamp.br 21

Respostas FTP

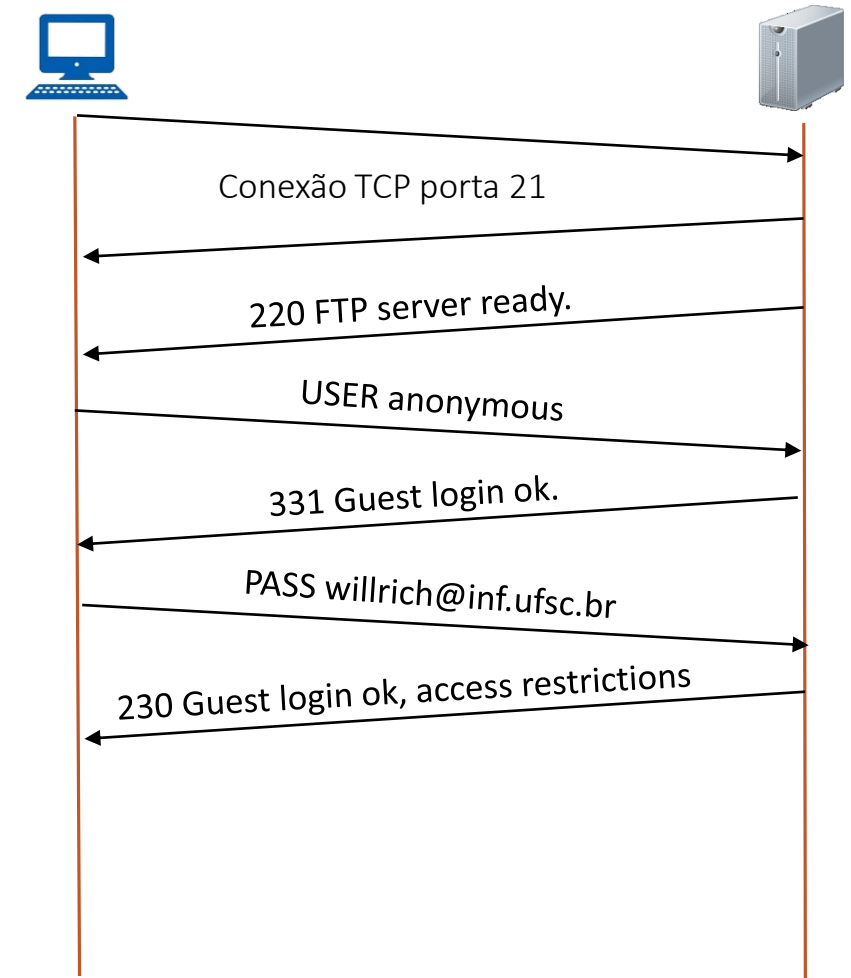
Enviadas sobre o canal de controle

- Linhas únicas contendo:
 - 3 dígitos de status
 - Mensagem de texto
- Primeiro dígito do status indica o tipo de resposta
 - '1': Resposta positiva preliminar (ok, mas espere)
 - '2': Resposta positiva (sucesso)
 - '3': Resposta positiva intermediária (esperando mais informações).
 - '4': Resposta negativa transiente (erro – tente novamente)
 - '5': Resposta negativa permanente (erro – não pode fazer).

FTP: Modo Ativo

Exemplo de conexão FTP modo ativo

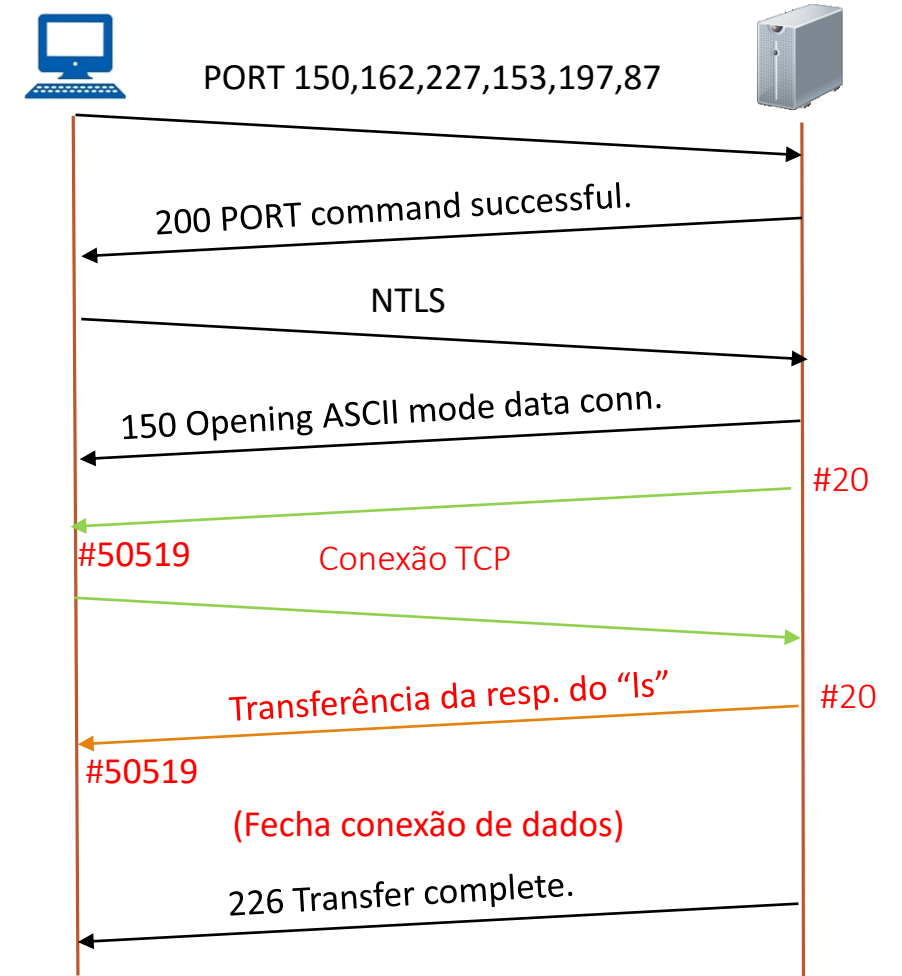
- Inicia conexão de controle FTP com servidor
- Servidor cumprimenta cliente
- Cliente faz o login (anônimo)
 - Note que a senha passar sem criptografia
- Agora cliente quer solicitar o conteúdo do diretório remoto
 - Necessário abrir conexão de dados antes



FTP: Modo Ativo

Exemplo de conexão FTP modo ativo

- Abrindo conexão de dados via comando PORT
 - Cliente informa ao servidor que ele quer receber conexão de dados no endereço IP 150.162.227.153 e na porta $197 \times 256 + 87 = 50519$
 - Servidor solicita conexão com o end. IP e porta informados
- Vendo o conteúdo do diretório
 - Comando NTLS
 - Conteúdo enviado na conexão de dados
 - Conexão de dados é fechada



Exemplo de FTP

The image displays a Wireshark packet capture window titled 'ftp1.pcap - Wireshark' with a filter set to 'tcp'. The packet list shows several TCP and FTP packets. A red oval highlights packets 22, 23, and 24, which represent the initial TCP connection setup. Packet 22 is a SYN from 192.168.0.197 to 128.113.126.13. Packet 23 is a SYN, ACK from 128.113.126.13 to 192.168.0.197. Packet 24 is an ACK from 192.168.0.197 to 128.113.126.13. A text box with the text 'Pedido de conexão TCP com a porta 21' points to these packets. Below the packet list, the packet details for packet 22 are expanded, showing the Ethernet II, Internet Protocol, and Transmission Control Protocol layers. A command prompt window is overlaid on the bottom right, showing the execution of the command 'ftp ftp.cs.rpi.edu'. The prompt shows the connection to the remote host, the login process, and the directory listing of the remote site.

Filter: tcp

No.	Time	Source	Destination	Protocol	Info
22	3.438	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [SYN] Seq=0 win=8192 Len=0 MSS=1460 WS=2 SACK_PERM=1
23	3.600	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [SYN, ACK] Seq=0 Ack=1 win=65535 Len=0 MSS=1380 WS=3
24	3.606	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=1 win=8192 Len=0
25	3.770	128.113.126.13	192.168.0.197	FTP	Response: 220 FTP server ready.
26	3.976	192.168.0.197	128.113.126.13	FTP	Request: USER anonymous
30	9.322	192.168.0.197	128.113.126.13	FTP	Request: PASS willrich@inf.ufsc.br
31	9.484	128.113.126.13	192.168.0.197	FTP	Response: 331 Guest login ok, send your email address as password.
32	9.686	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=17 Ack=82 win=8108 Len=0
33	13.381	192.168.0.197	128.113.126.13	FTP	Request: PASS willrich@inf.ufsc.br
34	13.611	128.113.126.13	192.168.0.197	FTP	Response: 230 Guest login ok, access restrictions apply.
35	13.821	192.168.0.197	128.113.126.13	FTP	Request: ls
36	14.521	192.168.0.197	128.113.126.13	FTP	Response: 200 PORT command successful.

Pedido de conexão TCP com a porta 21

```
C:\Users\willrich>ftp ftp.cs.rpi.edu
Conectado a valvet.cs.rpi.edu.
220 FTP server ready.
Usuário (valvet.cs.rpi.edu:(none)): anonymous
331 Guest login ok, send your email address as password.
Senha:
230 Guest login ok, access restrictions apply.
ftp> ls
200 PORT command successful.
150 Opening ASCII mode data connection for 'file list'.
.login_conf
etc
bin
welcome.msg
.pathmsg
.index.html
.snap
private
.notar
mirrors
pub
226 Transfer complete.
ftp: 97 bytes recebidos em 0.00Segundos 97000.00Kbytes/s.
ftp> cd pub
250 CWD command successful.
```

Exemplo de FTP

The image shows a Wireshark packet capture of an FTP session. The filter is set to 'tcp'. A red circle highlights packets 22, 23, and 24, which represent the TCP connection establishment. Packet 22 is a SYN from 192.168.0.197 to 128.113.126.13. Packet 23 is a SYN, ACK from 128.113.126.13 to 192.168.0.197. Packet 24 is an ACK from 192.168.0.197 to 128.113.126.13. A text box with the text 'Pedido de conexão TCP com a porta 21' is overlaid on the capture. The packet details for packet 22 are expanded, showing the Ethernet II, Internet Protocol, and Transmission Control Protocol layers. The TCP layer shows Src Port: taurus-wh (1610), Dst Port: ftp (21), Seq: 0, Len: 0. The packet bytes are also displayed at the bottom.

No.	Time	Source	Destination	Protocol	Info
22	3.438	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [SYN] Seq=0 win=8192 Len=0 MSS=1460 WS=2 SACK_PERM=1
23	3.600	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [SYN, ACK] Seq=0 Ack=1 win=65535 Len=0 MSS=1380 WS=3
24	3.606	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=1 win=8192 Len=0
25	3.770	128.113.126.13	192.168.0.197	FTP	Response: 220 FTP server ready.
26	3.976	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=17 Ack=82 win=8108 Len=0
30	9.322	192.168.0.197	128.113.126.13	FTP	Request: PASS willrich@inf.ufsc.br
31	9.484	128.113.126.13	192.168.0.197	FTP	Response: 331 Guest login ok, send your email address as password.
32	9.686	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=17 Ack=82 win=8108 Len=0
33	13.38	192.168.0.197	128.113.126.13	FTP	Request: PASS willrich@inf.ufsc.br
34	13.611	128.113.126.13	192.168.0.197	FTP	Response: 230 Guest login ok, access restrictions apply.
35	13.82	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=44 Ack=130 win=8060 Len=0
36	14.521	192.168.0.197	128.113.126.13	MSNMS	PNG

Frame 22: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)

Ethernet II, Src: IntelCor_2a:f0:71 (00:1f:3c:2a:f0:71), Dst: D-Link_c0:ce:df (00:1c:f0:c0:ce:df)

Internet Protocol, Src: 192.168.0.197 (192.168.0.197), Dst: 128.113.126.13 (128.113.126.13)

Transmission Control Protocol, Src Port: taurus-wh (1610), Dst Port: ftp (21), Seq: 0, Len: 0

```
0000  00 1c f0 c0 ce df 00 1f 3c 2a f0 71 08 00 45 00  ..... <*.q..E.
0010  00 34 4c 81 40 00 80 06 ee 56 c0 a8 00 c5 80 71  .4L.@... .V....q
0020  7e 0d 06 4a 00 15 55 94 00 bd 00 00 00 00 80 02  ~..J..U. ....
0030  20 00 32 7a 00 00 02 04 05 b4 01 03 03 02 01 01  .2z.... ....
0040  04 02  ..
```

File: "C:\Users\willrich\Documents\My Drop... Packets: 120 Displayed: 110 Marked: 0 Load time: 0:00.000 Profile: Default

Exemplo de FTP

The image shows a Wireshark packet capture of an FTP session. The filter is set to 'tcp'. The packet list shows several packets, with packet 25 highlighted in blue and circled in red. This packet is an FTP response from the server (128.113.126.13) to the client (192.168.0.197), indicating the server is ready (220 FTP server ready.). A red box with the text 'Mensagem de boas vindas do servidor' is overlaid on this packet. The packet details pane shows the structure of the frame, including Ethernet II, Internet Protocol, Transmission Control Protocol, and File Transfer Protocol (FTP). The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Info
22	3.438	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [SYN] Seq=0 win=8192 Len=0 MSS=1460 WS=2 SACK_PERM=1
23	3.600	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [SYN, ACK] Seq=0 Ack=1 win=65535 Len=0 MSS=1380 WS=3
24	3.600	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=1 win=8192 Len=0
25	3.770	128.113.126.13	192.168.0.197	FTP	Response: 220 FTP server ready.
26	3.978	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=24 win=8168 Len=0
30	9.322	192.168.0.197	128.113.126.13	FTP	Request: USER anonymous
31	9.484	128.113.126.13	192.168.0.197	FTP	Response: 331 Please send the email address as password.
32	9.686	192.168.0.197	128.113.126.13	FTP	Request: PASS willrich@inf.ufsc.br
33	13.381	192.168.0.197	128.113.126.13	FTP	Request: PASS willrich@inf.ufsc.br
34	13.611	128.113.126.13	192.168.0.197	FTP	Response: 230 Guest login ok, access restrictions apply.
35	13.821	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=44 Ack=130 win=8060 Len=0
36	14.571	192.168.0.197	128.113.126.13	MSNMS	PNG

Frame 25: 77 bytes on wire (616 bits), 77 bytes captured (616 bits)

Ethernet II, Src: D-Link_c0:ce:df (00:1c:f0:c0:ce:df), Dst: IntelCor_2a:f0:71 (00:1f:3c:2a:f0:71)

Internet Protocol, Src: 128.113.126.13 (128.113.126.13), Dst: 192.168.0.197 (192.168.0.197)

Transmission Control Protocol, Src Port: ftp (21), Dst Port: taurus-wh (1610), Seq: 1, Ack: 1, Len: 23

File Transfer Protocol (FTP)

220 FTP server ready.\r\n

0000 00 1f 3c 2a f0 71 00 1c f0 c0 ce df 08 00 45 00 ..<*.q..E.
0010 00 3f 84 45 40 00 2e 06 08 88 80 71 7e 0d c0 a8 .?.E@... ..q~...
0020 00 c5 00 15 06 4a da fa 15 d2 55 94 00 be 50 18J.. ..U...P.
0030 20 58 1b 84 00 00 32 32 30 20 46 54 50 20 73 65 X....22 0 FTP se
0040 72 76 65 72 20 72 65 61 64 79 2e 0d 0a rver rea dy...

File: "C:\Users\willrich\Documents\My Drop..." Packets: 120 Displayed: 110 Marked: 0 Load time: 0:00.000 Profile: Default

Exemplo de FTP

The image displays a Wireshark packet capture of an FTP session and a corresponding Windows command prompt window.

Wireshark Packet Capture:

- Filter:** tcp
- Table:**

No.	Time	Source	Destination	Protocol	Info
22	3.438	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [SYN] Seq=0 win=8192 Len=0 MSS=1460 WS=2 SACK_PERM=1
23	3.600	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [SYN, ACK] Seq=0 Ack=1 win=65535 Len=0 MSS=1380 WS=3
24	3.600	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=1 win=8192 Len=0
25	3.770	128.113.126.13	192.168.0.197	FTP	Response: 220 FTP server ready.
26	3.976	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=24 win=8168 Len=0
30	9.327	192.168.0.197	128.113.126.13	FTP	Request: USER anonymous
31	9.484	128.113.126.13	192.168.0.197	FTP	Response: 331 Guest login ok, send your email address as password.
32	9.686	192.168.0.197	128.113.126.13	FTP	Request: PASS willrich@inf.ufsc.br
33	13.381	192.168.0.197	128.113.126.13	FTP	Response: 230 Guest login ok, access restrictions apply.
34	13.611	128.113.126.13	192.168.0.197	FTP	Request: ls
35	13.821	192.168.0.197	128.113.126.13	FTP	Response: 200 PORT command successful.
36	14.571	192.168.0.197	128.113.126.13	FTP	Response: 150 Opening ASCII mode data connection for 'file list'.

Packet Details:

 - Frame 25: 77 bytes on wire (616 bits)
 - Ethernet II, Src: D-Link_C0:ce:00:00:00:00, Dst: 08:00:00:00:00:00
 - Internet Protocol, Src: 128.113.126.13, Dst: 192.168.0.197
 - Transmission Control Protocol, Src Port: 21, Dst Port: 21
 - File Transfer Protocol (FTP)
 - 220 FTP server ready. \r\n

Windows Command Prompt:

```
C:\Users\willrich>ftp ftp.cs.rpi.edu
Conectado a valvet.cs.rpi.edu.
220 FTP server ready.
Usuário (valvet.cs.rpi.edu:(none)): anonymous
331 Guest login ok, send your email address as password.
Senha:
230 Guest login ok, access restrictions apply.
ftp> ls
200 PORT command successful.
150 Opening ASCII mode data connection for 'file list'.
.login_conf
etc
bin
welcome.msg
.pathmsg
.index.html
.snap
private
.notar
mirrors
pub
226 Transfer complete.
ftp: 97 bytes recebidos em 0.00Segundos 97000.00Kbytes/s.
ftp> cd pub
250 CWD command successful.
```


Exemplo de FTP

The image shows a Wireshark capture of an FTP session. The filter is set to 'tcp'. The packet list shows several packets, with packet 30 highlighted in blue and circled in red. Packet 30 is an FTP 'Request: USER anonymous' from 192.168.0.197 to 128.113.126.13. A red box highlights the text 'Comando USER' over the packet details. The packet details pane shows the structure of the frame, including Ethernet II, Internet Protocol, Transmission Control Protocol, and File Transfer Protocol (FTP). The FTP section shows 'USER anonymous\r\n'. The packet bytes pane shows the raw data in hexadecimal and ASCII, with the ASCII column displaying '...e..US ER anony' and 'mous..'. The status bar at the bottom indicates 'Text item (text), 16 bytes'.

No.	Time	Source	Destination	Protocol	Info
22	3.438	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [SYN] Seq=0 win=8192 Len=0 MSS=1460 WS=2 SACK_PERM=1
23	3.600	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [SYN, ACK] Seq=0 Ack=1 win=65535 Len=0 MSS=1380 WS=3
24	3.600	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=1 win=8192 Len=0
25	3.770	128.113.126.13	192.168.0.197	FTP	Response: 220 FTP server ready.
26	3.976	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=24 win=8168 Len=0
30	9.327	192.168.0.197	128.113.126.13	FTP	Request: USER anonymous
31	9.484	128.113.126.13	192.168.0.197	FTP	Response: 331 Guest login ok, send your email address as password.
32	9.686	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=17 Ack=82 win=8108 Len=0
33	13.381	192.168.0.197	128.113.126.13	FTP	Request: PASS willrich@inf.ufsc.br
34	13.611	128.113.126.13	192.168.0.197	FTP	Response: 332 Password required for willrich@inf.ufsc.br. Restrictions apply. Password may be longer than 8060 Len=0
35	13.821	192.168.0.197	128.113.126.13	FTP	
36	14.571	192.168.0.197	128.113.126.13	FTP	

Comando USER

Frame 30: 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface
Ethernet II, Src: IntelCor_2a:f0:71 (00:1f:3c:2a:f0:71), Dst: D-Link_c0:ce:df (00:1c:f0:c0:ce:df)
Internet Protocol, Src: 192.168.0.197 (192.168.0.197), Dst: 128.113.126.13 (128.113.126.13)
Transmission Control Protocol, Src Port: taurus-wh (1610), Dst Port: ftp (21), Seq: 1, Ack: 24, Len: 16
File Transfer Protocol (FTP)
USER anonymous\r\n

0000 00 1c f0 c0 ce df 00 1f 3c 2a f0 71 08 00 45 00 <*.q..E.
0010 00 38 4c 84 40 00 80 06 ee 4f c0 a8 00 c5 80 71 .8L.@... .O.....q
0020 7e 0d 06 4a 00 15 55 94 00 be da fa 15 e9 50 18 ~..J..U.P.
0030 07 fa 12 65 00 00 55 53 45 52 20 61 6e 6f 6e 79 ...e..US ER anony
0040 6d 6f 75 73 0d 0a mous..

Text item (text), 16 bytes Packets: 120 Displayed: 110 Marked: 0 Load time: 0:00.000 Profile: Default

Exemplo de FTP

The image displays a Wireshark packet capture window titled 'ftp1.pcap - Wireshark' and a Windows Command Prompt window showing an FTP session.

Wireshark Packet Capture:

- Filter:** tcp
- Table:**

No.	Time	Source	Destination	Protocol	Info
22	3.438	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [SYN] Seq=0 win=8192 Len=0 MSS=1460 WS=2 SACK_PERM=1
23	3.600	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [SYN, ACK] Seq=0 Ack=1 win=65535 Len=0 MSS=1380 WS=3
24	3.600	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=1 win=8192 Len=0
25	3.770	128.113.126.13	192.168.0.197	FTP	Response: 220 FTP server ready.
26	3.976	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=24 win=8168 Len=0
30	9.327	192.168.0.197	128.113.126.13	FTP	Request: USER anonymous
31	9.484	128.113.126.13	192.168.0.197	FTP	Response: 331 Guest login ok, send your email address as password.
32	9.686	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=17 Ack=82 win=8108 Len=0
33	13.381	192.168.0.197	128.113.126.13	FTP	Request: PASS willrich@inf.ufsc.br
34	13.611	128.113.126.13	192.168.0.197	FTP	Response: 332 Password accepted. Restrictions apply.
35	13.821	192.168.0.197	128.113.126.13	FTP	Request: CWD /
36	14.571	192.168.0.197	128.113.126.13	FTP	Response: 200 CWD successful.

Command Prompt:

```
C:\Users\willrich>ftp ftp.cs.rpi.edu
Conectado a valvet.cs.rpi.edu.
220 FTP server ready.
Usuário (valvet.cs.rpi.edu:(none)): anonymous
331 Guest login ok, send your email address as password.
Senha:
230 Guest login ok, access restrictions apply.
ftp> ls
200 PORT command successful.
150 Opening ASCII mode data connection for 'file list'.
.login_conf
etc
bin
welcome.msg
.pathmsg
.index.html
.snap
private
.notar
mirrors
pub
```

Exemplo de FTP

ftp1.pcap - Wireshark

Filter: tcp

No.	Time	Source	Destination	Protocol	Info
22	3.438	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [SYN] Seq=0 win=8192 Len=0 MSS=1460 WS=2 SACK_PERM=1
23	3.600	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [SYN, ACK] Seq=0 Ack=1 win=65535 Len=0 MSS=1380 WS=3
24	3.600	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=1 win=8192 Len=0
25	3.770	128.113.126.13	192.168.0.197	FTP	Response: 220 FTP server ready.
26	3.976	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=24 win=8168 Len=0
30	9.322	192.168.0.197	128.113.126.13	FTP	Request: USER anonymous
31	9.484	128.113.126.13	192.168.0.197	FTP	Response: 331 Guest login ok, send your email address as password.
32	9.686	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=17 Ack=82 win=8108 Len=0
33	13.38	192.168.0.197	128.113.126.13	FTP	Request: PASS willrich@inf.ufsc.br
34	13.611	128.113.126.13	192.168.0.197	FTP	Response: 230 Guest login ok, access restrictions apply.
35	13.821	192.168.0.197			
36	14.521	192.168.0.197			

Resposta do comando USER

Frame 31: 112 bytes on wire (896 bits), 112 bytes captured (896 bits)

Ethernet II, Src: D-Link_c0:ce:df (00:1c:f0:c0:ce:df), Dst: IntelCor_2a:f0:71 (00:1f:3c:2a:f0:71)

Internet Protocol, Src: 128.113.126.13 (128.113.126.13), Dst: 192.168.0.197 (192.168.0.197)

Transmission Control Protocol, Src Port: ftp (21), Dst Port: taurus-wh (1610), Seq: 24, Ack: 17, Len: 58

File Transfer Protocol (FTP)

331 Guest login ok, send your email address as password.\r\n

0000 00 1f 3c 2a f0 71 00 1c f0 c0 ce df 08 00 45 00 ..<*.q..E.
0010 00 62 84 7b 40 00 2e 06 08 2f 80 71 7e 0d c0 a8 .b.{@... ./..q~...
0020 00 c5 00 15 06 4a da fa 15 e9 55 94 00 ce 50 18J.. ..U...P.
0030 20 58 32 14 00 00 33 33 31 20 47 75 65 73 74 20 x2...33 1 Guest
0040 6c 6f 67 69 6e 20 6f 6b 2c 20 73 65 6e 64 20 79 login ok , send y
0050 6f 75 72 20 65 6d 61 69 6c 20 61 64 64 72 65 73 our email address
0060 73 20 61 73 20 70 61 73 73 77 6f 72 64 2e 0d 0a s as password...

File: "C:\Users\willrich\Documents\My Drop..." Packets: 120 Displayed: 110 Marked: 0 Load time: 0:00.000 Profile: Default

Exemplo de FTP

The image displays a Wireshark packet capture window titled 'ftp1.pcap - Wireshark' and a Windows Command Prompt window showing an FTP session.

Wireshark Packet Capture:

- Filter: tcp
- Table with columns: No., Time, Source, Destination, Protocol, Info.
- Packet 31 is highlighted with a red circle. Its info field is: `Response: 331 Guest login ok, send your email address as password.`
- Packet 32 is highlighted with a red circle. Its info field is: `taurus-wh > ftp [ACK] Seq=17 Ack=82 win=8108 Len=0`
- Packet 33 is highlighted with a red circle. Its info field is: `Request: PASS willrich@inf.ufsc.br`
- Packet 34 is highlighted with a red circle. Its info field is: `Response: 230 Guest login ok, access restrictions apply.`

Command Prompt:

```
C:\Users\willrich>ftp ftp.cs.rpi.edu
Conectado a valvet.cs.rpi.edu.
220 FTP server ready.
Usuário (valvet.cs.rpi.edu:(none)): anonymous
331 Guest login ok, send your email address as password.
Senha:
230 Guest login ok, access restrictions apply.
ftp> ls
200 PORT command successful.
150 Opening ASCII mode data connection for 'file list'.
.login_conf
etc
bin
welcome.msg
.pathmsg
.index.html
.snap
private
.notar
mirrors
pub
```

A red circle highlights the prompt `Usuário (valvet.cs.rpi.edu:(none)): anonymous` and the response `331 Guest login ok, send your email address as password.` in the command prompt.

Exemplo de FTP

The image shows a Wireshark capture of an FTP session. The filter is set to 'tcp'. The packet list shows several packets, with packet 33 highlighted in blue. A red oval highlights packet 33, and a red box with the text 'Comando USER' is placed over it. The packet details pane shows the structure of packet 33, which is an FTP 'PASS' command. The packet bytes pane shows the raw data of packet 33, which is the ASCII string 'PASS willrich@inf.ufsc.br\r\n'.

No.	Time	Source	Destination	Protocol	Info
22	3.438	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [SYN] Seq=0 win=8192 Len=0 MSS=1460 WS=2 SACK_PERM=1
23	3.600	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [SYN, ACK] Seq=0 Ack=1 win=65535 Len=0 MSS=1380 WS=3
24	3.600	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=1 win=8192 Len=0
25	3.770	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [ACK] Seq=1 Ack=1 win=8192 Len=0
26	3.976	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=1 win=8192 Len=0
30	9.322	192.168.0.197	128.113.126.13	FTP	Request: USER anonymous
31	9.484	128.113.126.13	192.168.0.197	FTP	Response: 331 Guest login ok, send your email address as password.
32	9.686	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=17 Ack=82 win=8108 Len=0
33	13.381	192.168.0.197	128.113.126.13	FTP	Request: PASS willrich@inf.ufsc.br
34	13.611	128.113.126.13	192.168.0.197	FTP	Response: 230 Guest login ok, access restrictions apply.
35	13.821	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=44 Ack=130 win=8060 Len=0
36	14.571	192.168.0.197	128.113.126.13	MSNMS	PNG

Frame 33: 81 bytes on wire (648 bits), 81 bytes captured (648 bits)

Ethernet II, Src: IntelCor_2a:f0:71 (00:1f:3c:2a:f0:71), Dst: D-Link_c0:ce:df (00:1c:f0:c0:ce:df)

Internet Protocol, Src: 192.168.0.197 (192.168.0.197), Dst: 128.113.126.13 (128.113.126.13)

Transmission Control Protocol, Src Port: taurus-wh (1610), Dst Port: ftp (21), Seq: 17, Ack: 82, Len: 27

File Transfer Protocol (FTP)

PASS willrich@inf.ufsc.br\r\n

0000 00 1c f0 c0 ce df 00 1f 3c 2a f0 71 08 00 45 00 <*.q..E.
0010 00 43 4c 86 40 00 80 06 ee 42 c0 a8 00 c5 80 71 .CL.@... .B.....q
0020 7e 0d 06 4a 00 15 55 94 00 ce da fa 16 23 50 18 ~..J..U.#P.
0030 07 eb cb 96 00 00 50 41 53 53 20 77 69 6c 6c 72PA SS willr
0040 69 63 68 40 69 6e 66 2e 75 66 73 63 2e 62 72 0d ich@inf. ufsc.br.
0050 0a .

File: "C:\Users\willrich\Documents\My Drop..." Packets: 120 Displayed: 110 Marked: 0 Load time: 0:00.000 Profile: Default

Exemplo de FTP

The image displays a Wireshark packet capture of an FTP session and a terminal window showing the command prompt.

Wireshark Packet Capture:

- Filter: tcp
- Table with columns: No., Time, Source, Destination, Protocol, Info.
- Packet 33 is highlighted in blue and circled in red. It is an FTP Request: PASS willrich@inf.ufsc.br.
- Packet 34 is an FTP Response: 230 Guest login ok, access restrictions apply.
- Packet 35 is a TCP ACK: taurus-wh > ftp [ACK] Seq=44 Ack=130 win=8060 Len=0.

Terminal Window (Prompt de Comando):

```
C:\Users\willrich>ftp ftp.cs.rpi.edu
Conectado a valvet.cs.rpi.edu.
220 FTP server ready.
Usuário (valvet.cs.rpi.edu:(none)): anonymous
331 Guest login ok, send your email address as password.
Senha:
230 Guest login ok, access restrictions apply.
ftp> ls
200 PORT command successful.
150 Opening ASCII mode data connection for 'file list'.
.login_conf
etc
bin
welcome.msg
.pathmsg
.index.html
.snap
private
.notar
mirrors
pub
```

Exemplo de FTP

Wireshark packet capture of an FTP session. The packet list shows the following details:

No.	Time	Source	Destination	Protocol	Info
22	3.438	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [SYN] Seq=0 win=8192 Len=0 MSS=1460 WS=2 SACK_PERM=1
23	3.600	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [SYN, ACK] Seq=0 Ack=1 win=65535 Len=0 MSS=1380 WS=3
24	3.600	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=1 Ack=1 win=8192 Len=0
25	3.770	128.113.126.13	192.168.0.197	FTP	Response: 220 FTP server ready
26	3.976	192.168.0.197	128.113.126.13	FTP	Request: USER willrich@inf.ufsc.br Len=0
30	9.322	192.168.0.197	128.113.126.13	FTP	Response: 331 Guest login ok, send your email address as password.
31	9.484	128.113.126.13	192.168.0.197	FTP	Request: PASS willrich@inf.ufsc.br
32	9.686	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=17 Ack=82 win=8108 Len=0
33	13.381	192.168.0.197	128.113.126.13	FTP	Request: PASS willrich@inf.ufsc.br
34	13.611	128.113.126.13	192.168.0.197	FTP	Response: 230 Guest login ok, access restrictions apply.
35	13.821	192.168.0.197	128.113.126.13	TCP	taurus-wh > ftp [ACK] Seq=44 Ack=130 win=8060 Len=0
36	14.571	192.168.0.197	128.113.126.13	MSNMS	PNG

Packet 34 details:

- Frame 34: 102 bytes on wire (816 bits), 102 bytes captured (816 bits)
- Ethernet II, Src: D-Link_c0:ce:df (00:1c:f0:c0:ce:df), Dst: IntelCor_2a:f0:71 (00:1f:3c:2a:f0:71)
- Internet Protocol, Src: 128.113.126.13 (128.113.126.13), Dst: 192.168.0.197 (192.168.0.197)
- Transmission Control Protocol, Src Port: ftp (21), Dst Port: taurus-wh (1610), Seq: 82, Ack: 44, Len: 48
- File Transfer Protocol (FTP)
 - 230 Guest login ok, access restrictions apply.\r\n

Packet 34 hex dump:

```
0000  00 1f 3c 2a f0 71 00 1c f0 c0 ce df 08 00 45 00  ..<*.q.. .....E.
0010  00 58 84 a1 40 00 2e 06 08 13 80 71 7e 0d c0 a8  .X..@... ..q~...
0020  00 c5 00 15 06 4a da fa 16 23 55 94 00 e9 50 18  .....J.. .#U...P.
0030  20 58 d5 77 00 00 32 33 30 20 47 75 65 73 74 20  X.w..23 0 Guest
0040  6c 6f 67 69 6e 20 6f 6b 2c 20 61 63 63 65 73 73  login ok , access
0050  20 72 65 73 74 72 69 63 74 69 6f 6e 73 20 61 70  restric tions ap
0060  70 6c 79 2e 0d 0a                                ply...
```

Exemplo de FTP

The image shows a Wireshark packet capture window titled 'ftp1.pcap - Wireshark'. The filter is set to 'tcp'. A red circle highlights packets 39, 40, and 41. Packet 39 is an FTP Request: PORT 192,168,0,197,6,75. Packet 40 is an FTP Response: 200 PORT command successful. Packet 41 is an FTP Request: NLST. A red oval highlights the source and destination IP addresses in packets 39 and 40. A text box is overlaid on the packet list, containing the following information:

Comando Port:
IP: 192.168.0.197
Porta Y: $6 \times 256 + 75 = 1611$

The packet details pane shows the following information for packet 39:

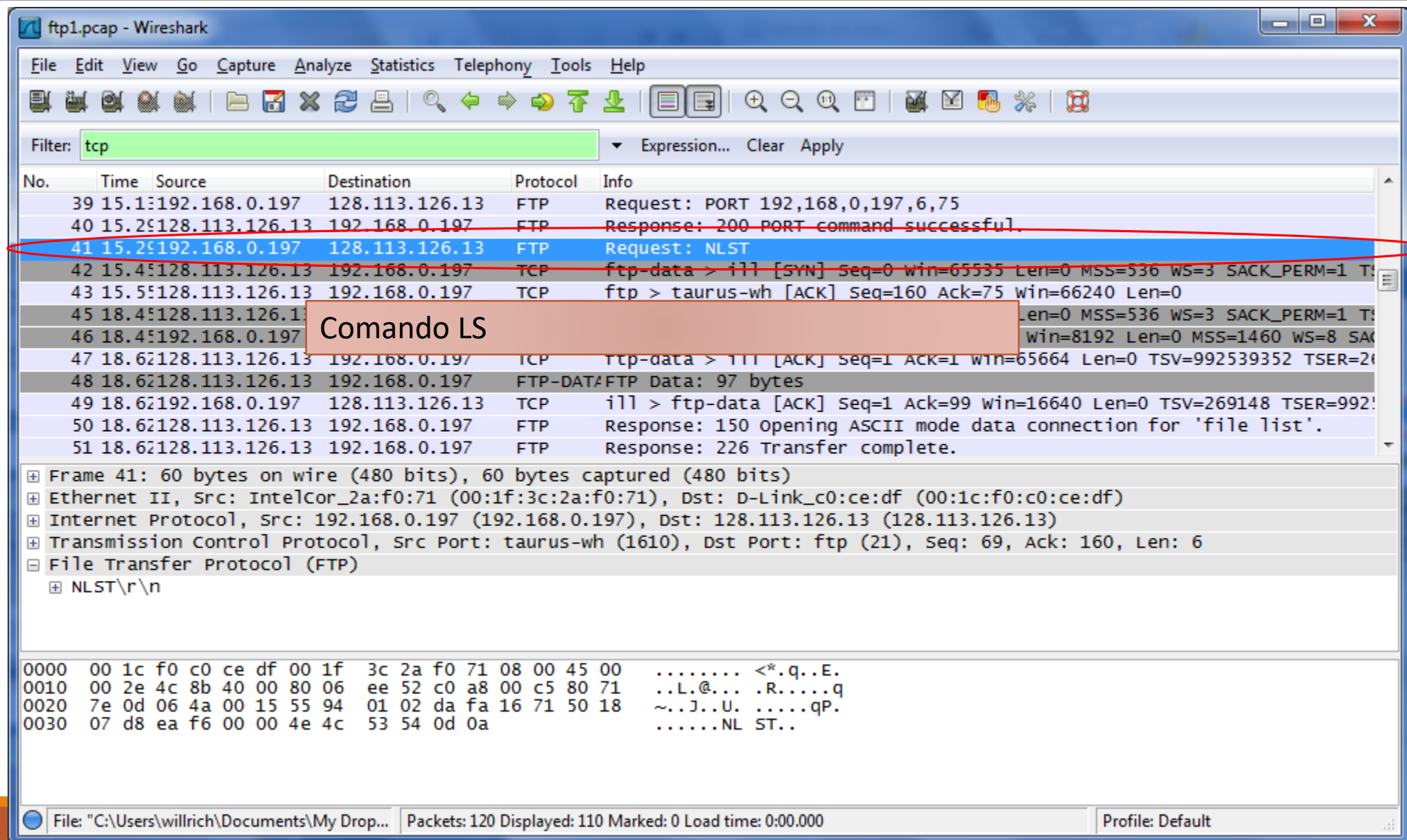
- Frame 39: 79 bytes on wire (632 bits), 79 bytes captured (632 bits)
- Ethernet II, Src: IntelCor_2a:f0:71 (00:1f:3c:2a:f0:71), Dst: D-Link_c0:ce:df (00:1c:f0:c0:ce:df)
- Internet Protocol, Src: 192.168.0.197 (192.168.0.197), Dst: 128.113.126.13 (128.113.126.13)
- Transmission Control Protocol, Src Port: taurus-wh (1610), Dst Port: ftp (21), Seq: 44, Ack: 130, Len: 25
- File Transfer Protocol (FTP)
 - PORT 192,168,0,197,6,75\r\n

The packet bytes pane shows the raw data for packet 39:

```
0000  00 1c f0 c0 ce df 00 1f 3c 2a f0 71 08 00 45 00  ..... <*.q..E.
0010  00 41 4c 8a 40 00 80 06 ee 40 c0 a8 00 c5 80 71  .AL.@... .@.....q
0020  7e 0d 06 4a 00 15 55 94 00 e9 da fa 16 53 50 18  ~..J..U. ....SP.
0030  07 df 12 3b 00 00 50 4f 52 54 20 31 39 32 2c 31  ...;...PO RT 192,1
0040  36 38 2c 30 2c 31 39 37 2c 36 2c 37 35 0d 0a    68,0,197 ,6,75..
```

The status bar at the bottom shows: File: "C:\Users\willrich\Documents\My Drop... Packets: 120 Displayed: 110 Marked: 0 Load time: 0:00.000 Profile: Default

Exemplo de FTP



The image shows a Wireshark packet capture window titled "ftp1.pcap - Wireshark". The filter is set to "tcp". The packet list shows several packets, with packet 41 highlighted in blue and circled in red. Packet 41 is an FTP Request: NLST. Below the packet list, the packet details for packet 41 are expanded, showing the File Transfer Protocol (FTP) section with the command "NLST\r\n".

No.	Time	Source	Destination	Protocol	Info
39	15.13	192.168.0.197	128.113.126.13	FTP	Request: PORT 192,168,0,197,6,75
40	15.25	128.113.126.13	192.168.0.197	FTP	Response: 200 PORT command successful.
41	15.25	192.168.0.197	128.113.126.13	FTP	Request: NLST
42	15.45	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [SYN] Seq=0 win=65535 Len=0 MSS=536 WS=3 SACK_PERM=1 TS...
43	15.55	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [ACK] Seq=160 Ack=75 win=66240 Len=0
45	18.45	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [ACK] Seq=1 ACK=1 win=65664 Len=0 TSV=992539352 TSER=2...
46	18.45	192.168.0.197	128.113.126.13	FTP	Response: 150 Opening ASCII mode data connection for 'file list'.
47	18.62	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [ACK] Seq=1 ACK=1 win=65664 Len=0 TSV=992539352 TSER=2...
48	18.62	128.113.126.13	192.168.0.197	FTP-DATA	FTP Data: 97 bytes
49	18.62	192.168.0.197	128.113.126.13	TCP	ill > ftp-data [ACK] Seq=1 Ack=99 win=16640 Len=0 TSV=269148 TSER=992...
50	18.62	128.113.126.13	192.168.0.197	FTP	Response: 150 Opening ASCII mode data connection for 'file list'.
51	18.62	128.113.126.13	192.168.0.197	FTP	Response: 226 Transfer complete.

Comando LS

Frame 41: 60 bytes on wire (480 bits), 60 bytes captured (480 bits)
Ethernet II, Src: IntelCor_2a:f0:71 (00:1f:3c:2a:f0:71), Dst: D-Link_c0:ce:df (00:1c:f0:c0:ce:df)
Internet Protocol, Src: 192.168.0.197 (192.168.0.197), Dst: 128.113.126.13 (128.113.126.13)
Transmission Control Protocol, Src Port: taurus-wh (1610), Dst Port: ftp (21), Seq: 69, Ack: 160, Len: 6
File Transfer Protocol (FTP)
NLST\r\n

0000 00 1c f0 c0 ce df 00 1f 3c 2a f0 71 08 00 45 00 <*.q..E.
0010 00 2e 4c 8b 40 00 80 06 ee 52 c0 a8 00 c5 80 71 ..L.@... .R.....q
0020 7e 0d 06 4a 00 15 55 94 01 02 da fa 16 71 50 18 ~..J..U.qP.
0030 07 d8 ea f6 00 00 4e 4c 53 54 0d 0aNL ST..

File: "C:\Users\willrich\Documents\My Drop... Packets: 120 Displayed: 110 Marked: 0 Load time: 0:00.000 Profile: Default

Exemplo de FTP

The image shows a Wireshark capture of an FTP session. The filter is set to 'tcp'. The packet list shows several packets, with packet 42 highlighted. A red circle highlights packets 42, 43, 45, 46, and 47. A text box is overlaid on the packet details for packet 42, stating: "Estabelece conexão TCP com porta origem 20 (FTP-DATA) e destino 1611". The packet details for packet 42 show: "Transmission Control Protocol, Src Port: ftp-data (20), Dst Port: ill (1611), Seq: 0, Len: 0". The packet bytes pane shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Info
39	15.13	192.168.0.197	128.113.126.13	FTP	Request: PORT 192,168,0,197,6,75
40	15.25	128.113.126.13	192.168.0.197	FTP	Response: 200 PORT command successful.
41	15.25	192.168.0.197	128.113.126.13	FTP	Request: NLST
42	15.45	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [SYN] Seq=0 win=65535 Len=0 MSS=536 WS=3 SACK_PERM=1 TSV=992539352 TSER=269148
43	15.55	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [ACK] Seq=160 Ack=75 win=66240 Len=0
45	18.45	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [SYN] Seq=0 win=65535 Len=0 MSS=536 WS=3 SACK_PERM=1 TSV=992539352 TSER=269148
46	18.45	192.168.0.197	128.113.126.13	TCP	ill > ftp-data [SYN, ACK] Seq=0 Ack=1 win=8192 Len=0 MSS=1460 WS=8 SACK_PERM=1 TSV=269148 TSER=992539352
47	18.62	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [ACK] Seq=1 Ack=1 win=65664 Len=0 TSV=992539352 TSER=269148
48	18.62	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [ACK] Seq=1 Ack=1 win=65664 Len=0 TSV=992539352 TSER=269148
49	18.62	192.168.0.197	128.113.126.13	TCP	ill > ftp-data [ACK] Seq=0 Ack=1 win=8192 Len=0 MSS=1460 WS=8 SACK_PERM=1 TSV=269148 TSER=992539352
50	18.62	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [ACK] Seq=1 Ack=1 win=65664 Len=0 TSV=992539352 TSER=269148
51	18.62	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [ACK] Seq=1 Ack=1 win=65664 Len=0 TSV=992539352 TSER=269148

Estabelece conexão TCP com porta origem 20 (FTP-DATA) e destino 1611

Frame 42: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
Ethernet II, Src: D-Link_c0:ce:df (00:1c:f0:c0:ce:df), Dst: IntelCor_2a:f0:71 (00:1f:3c:2a:f0:71)
Internet Protocol, Src: 128.113.126.13 (128.113.126.13), Dst: 192.168.0.197 (192.168.0.197)
Transmission Control Protocol, Src Port: ftp-data (20), Dst Port: ill (1611), Seq: 0, Len: 0

0000 00 1f 3c 2a f0 71 00 1c f0 c0 ce df 08 00 45 00 ..<*.q..E.
0010 00 3c 84 af 40 00 2e 06 08 21 80 71 7e 0d c0 a8 <..@... !.q~...
0020 00 c5 00 14 06 4b 96 b3 47 81 00 00 00 00 a0 02K.. G.....
0030 ff ff 85 7c 00 00 02 04 02 18 01 03 03 03 04 02 ...|.
0040 08 0a 3b 28 e6 7b 00 00 00 00 ..;(.{. ..

File: "C:\Users\willrich\Documents\My Drop... Packets: 120 Displayed: 110 Marked: 0 Load time: 0:00.000 Profile: Default

Exemplo de FTP

The image shows a Wireshark capture of an FTP session. The packet list on the left shows several packets, with packet 48 highlighted in blue. A red circle is drawn around packet 48 and packet 49. A red box is drawn around the packet details for packet 48, which is labeled "Resposta do ls".

Filter: tcp

No.	Time	Source	Destination	Protocol	Info
43	15.55	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [ACK] Seq=160 Ack=75 win=66240 Len=0
45	18.45	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [SYN] Seq=0 win=65535 Len=0 MSS=536 WS=3 SACK_PERM=1
46	18.45	192.168.0.197	128.113.126.13	TCP	ill > ftp-data [SYN, ACK] Seq=0 Ack=1 win=8192 Len=0 MSS=1460 WS=8
47	18.62	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [ACK] Seq=1 Ack=1 win=65664 Len=0 TSV=992539352 TSER
48	18.62	128.113.126.13	192.168.0.197	FTP-DATA	FTP Data: 97 bytes
49	18.62	192.168.0.197	128.113.126.13	TCP	ill > ftp-data [ACK] Seq=1 Ack=99 win=16640 Len=0 TSV=269148 TSER=9
50	18.62	128.113.126.13	192.168.0.197	FTP	Response: 150 Opening ASCII mode data connection for 'file list'.
51	18.62	128.113.126.13	192.168.0.197	FTP	Response: 226 Transfer complete.

Frame 48: 163 bytes on wire (1304 bits) captured (163 bytes) over Ethernet II
Ethernet II, Src: D... (08:00:27:3c:2a:f0:71), Dst: 192.168.0.197 (08:00:27:3c:2a:f0:71)
Internet Protocol, Src: 128.113.126.13 (128.113.126.13), Dst: 192.168.0.197 (192.168.0.197)
Transmission Control Protocol, Src Port: ftp-data (20), Dst Port: ill (1611), Seq: 1, Ack: 1, Len: 97
FTP Data

Resposta do ls

```
0000 00 1f 3c 2a f0 71 00 1c f0 c0 ce df 08 00 45 00 ..<*.q.. .....E.
0010 00 95 84 c0 40 00 2e 06 07 b7 80 71 7e 0d c0 a8 ....@... ..q~...
0020 00 c5 00 14 06 4b 96 b3 47 82 1e 5c f5 db 80 19 .....K.. G.. \....
0030 20 10 14 6f 00 00 01 01 08 0a 3b 28 f2 d8 00 04 ..o.... .;(. ....
0040 1b 4b 2e 6c 6f 67 69 6e 5f 63 6f 6e 66 0d 0a 65 .K.login _conf..e
0050 74 63 0d 0a 62 69 6e 0d 0a 77 65 6c 63 6f 6d 65 tc..bin. .welcome
0060 2e 6d 73 67 0d 0a 2e 70 61 74 68 6d 73 67 0d 0a .msg...p athmsg..
0070 2e 69 6e 64 65 78 2e 68 74 6d 6c 0d 0a 2e 73 6e .index.h tml...sn
0080 61 70 0d 0a 70 72 69 76 61 74 65 0d 0a 2e 6e 6f ap..priv ate...no
0090 74 61 72 0d 0a 6d 69 72 72 6f 72 73 0d 0a 70 75 tar..mir rors..pu
00a0 62 0d 0a b..
```

File: "C:\Users\willrich\Documents\My Drop... Packets: 120 Displayed: 110 Marked: 0 Load time: 0:00.000 Profile: Default

Exemplo de FTP

The image displays a network packet capture in Wireshark and a corresponding Windows command prompt window, illustrating an FTP session.

Wireshark Packet Capture:

- Filter:** tcp
- Table:**

No.	Time	Source	Destination	Protocol	Info
43	15.5	128.113.126.13	192.168.0.197	TCP	ftp > taurus-wh [ACK] Seq=160 Ack=75 win=66240 Len=0
45	18.4	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [SYN] Seq=0 win=65535 Len=0 MSS=536 WS=3 SACK_PERM=1
46	18.4	192.168.0.197	128.113.126.13	TCP	ill > ftp-data [SYN, ACK] Seq=0 Ack=1 win=8192 Len=0 MSS=1460 WS=8
47	18.6	128.113.126.13	192.168.0.197	TCP	ftp-data > ill [ACK] Seq=1 Ack=1 win=65664 Len=0 TSV=992539352 TSER
48	18.6	128.113.126.13	192.168.0.197	FTP-DATA	FTP Data: 97 bytes
49	18.6	192.168.0.197	128.113.126.13	TCP	ill > ftp-data [ACK] Seq=1 Ack=99 win=16640 Len=0 TSV=269148 TSER=9
50	18.6	128.113.126.13	192.168.0.197	FTP	Response: 150 Opening ASCII mode data connection for 'file list'.
51	18.6	128.113.126.13	192.168.0.197	FTP	Response: 226 Transfer complete.

- Packet 48 Detail:**
 - Frame 48: 163 bytes
 - Ethernet II, Src: D-LINK_CV
 - Internet Protocol, Src: 128.113.126.13
 - Transmission Control Protocol
 - FTP Data

Windows Command Prompt:

```
C:\Users\willrich>ftp ftp.cs.rpi.edu
Conectado a valvet.cs.rpi.edu.
220 FTP server ready.
Usuário (valvet.cs.rpi.edu:(none)): anonymous
331 Guest login ok, send your email address as password.
Senha:
230 Guest login ok, access restrictions apply.
ftp> ls
200 PORT command successful.
150 Opening ASCII mode data connection for 'file list'.
.login_conf
etc
bin
welcome.msg
.pathmsg
.index.html
.snap
private
.notar
mirrors
pub
226 Transfer complete.
ftp: 97 bytes recebidos em 0,00Segundos 97000,00Kbytes/s.
ftp> cd pub
250 CWD command successful.
```

Redacted Area: A red box labeled "Resposta do ls" (ls response) covers the FTP data packet (No. 48) in the Wireshark packet list.

FTP: Modo Ativo

Geralmente não funciona: Firewalls bloqueiam pedidos de conexão vinda de fora da rede local

- Cliente abre o canal de comandos da porta cliente
- Para transferência de dados, cliente envia comando PORT indicando porta para receber dados
- Servidor cria conexão de dados (bloqueada pelo firewall)



FTP: Modo Ativo

```
Prompt de Comando - ftp ftp.unicamp.br

C:\Users\willrich>ftp ftp.unicamp.br
Conectado a ftp.unicamp.br.
220 FtpServer da Unicamp
200 UTF8 set to on
Usuário (ftp.unicamp.br:(none)): anonymous
331 Anonymous login ok, send your complete email address as your pass
Senha:
230-
    Bem-vindo ao servidor FTP da Unicamp!
    Sugestoes e comentarios, por favor entre em contato
    com <apoio@ccuec.unicamp.br>.

----

Welcome to UNICAMP's FTP server!
For comments on this site, please contact <apoio@ccuec.unicamp.br>.

You are user (4) of (300) simultaneous users allowed.

230 Anonymous access granted, restrictions apply
ftp> ls
200 PORT command successful
425 Unable to build data connection: Operation timed out
ftp> █
```

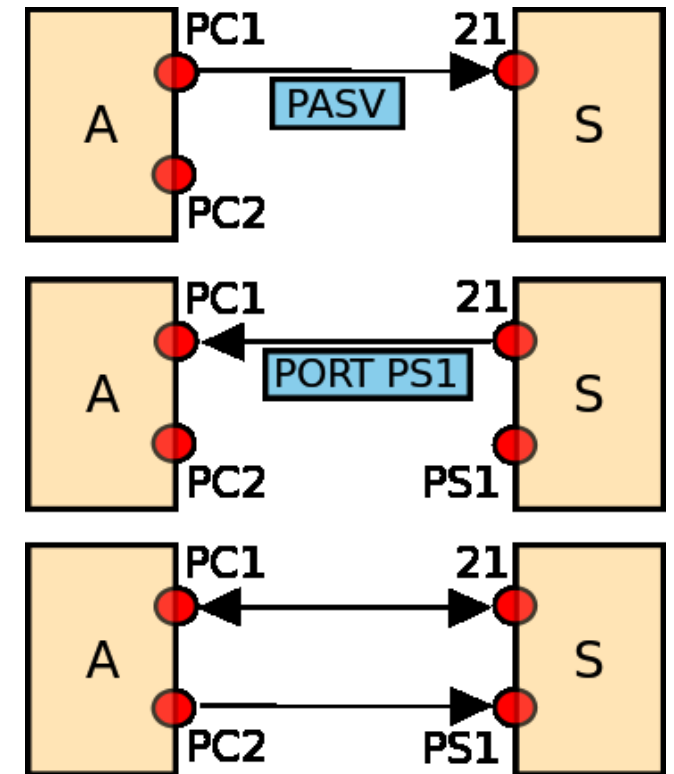
FTP Ativo (com Firewall)



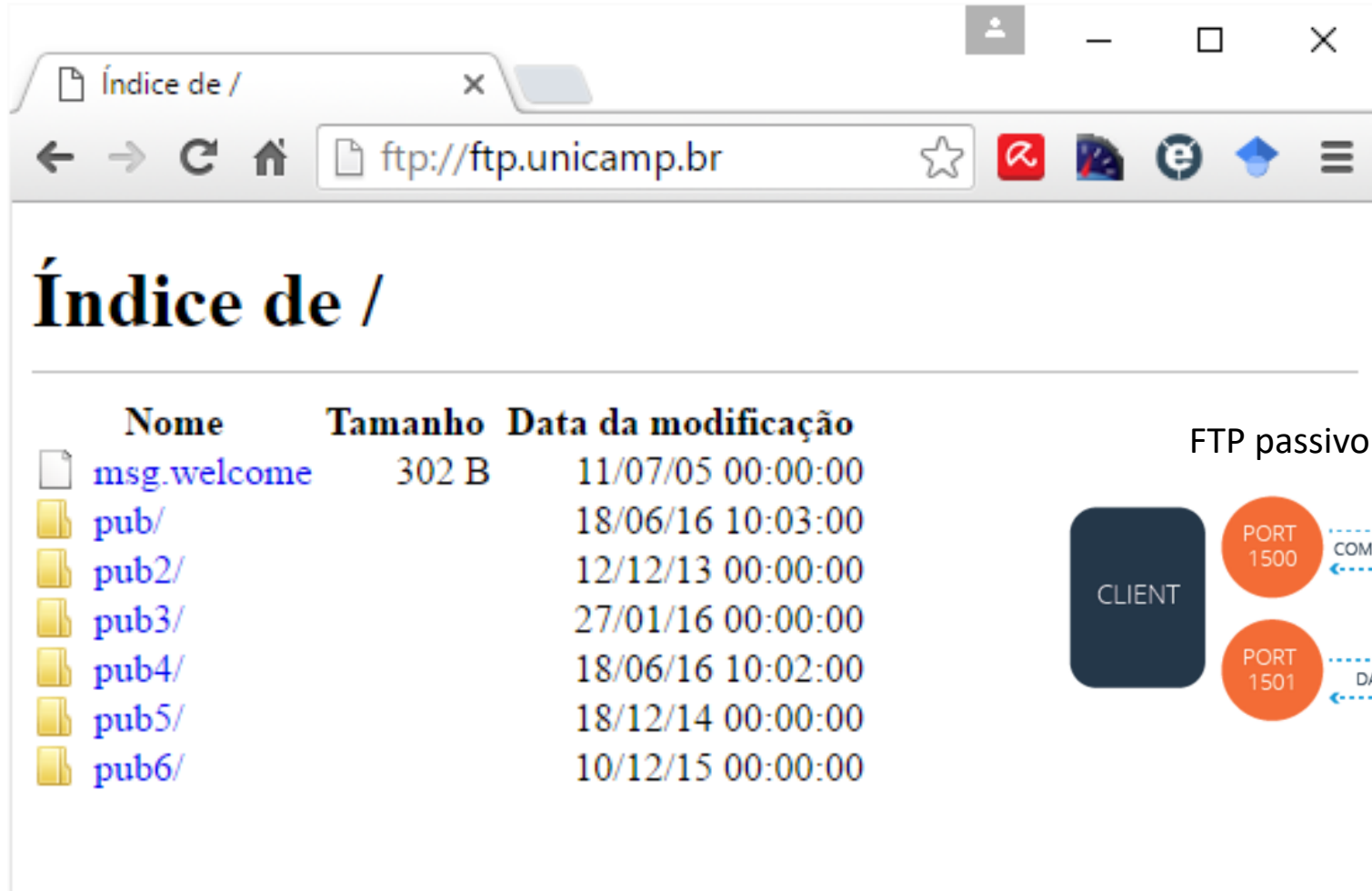
FTP: Modo Passivo

Operação

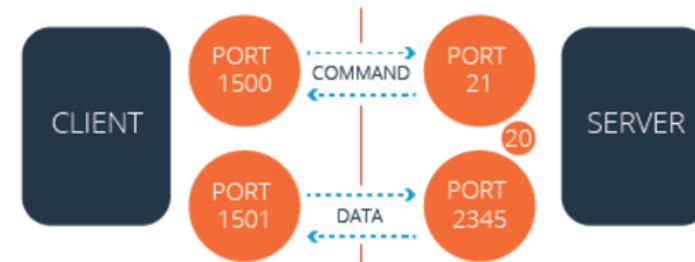
- Cliente abre o canal de comandos
- Cliente envia o comando “PASV” para o servidor através do canal de comandos
 - Esse comando muda a transmissão para o modo passivo;
 - Servidor responde através do canal de comandos a porta que irá escutar para o canal de dados (PS1);
- Cliente abre o canal de dados para a porta PS1 do servidor



FTP: Modo passivo



FTP passivo (com Firewall)



Características do FTP

Autenticação FTP

- Para usar o FTP é preciso ter uma conta (login) no host onde o servidor está rodando
- Servidores FTP possuem um usuário padrão chamado “anonymous”. Quando habilitado, usuários não precisam ter uma conta naquele host
- Usuário “anonymous” não tem senha: o servidor pede normalmente o e-mail do usuário (mas não checa)

Características do FTP

Desvantagens do FTP

- Passwords e conteúdo de arquivos trafegam na rede de forma “plana” (não criptografados)
- Negociação da “porta de dados” entre cliente e servidor no modo ativo
 - dificulta o trabalho de configuração de firewall
- No modo ativo é possível dizer ao servidor para estabelecer a “conexão de dados” com um terceiro host, indicando IP e porta

FTP Seguro

FTPS

- Uma extensão do FTP que adiciona suporte ao TLS (Transport Layer Security), e anteriormente SSL (proibido pela [RFC7568](#))
 - TLS permite a certificação do servidor e criptografia dos dados
 - Modo implícito: usa conexão segura desde o início
 - Modo explícito: conexão FTP inicia aberta, e o cliente pede explicitamente segurança

FTP Seguro

SFTP (SSH FTP)

- Não é o protocolo FTP seguro!
- Um protocolo de rede que fornece acesso, transferência e gerenciamento de arquivos sobre um canal confiável.
- Um protocolo binário e usa apenas uma conexão (não separa conexão de comando e dados)
- É tecnologicamente superior ao FTPS

Pontos Importantes

FTP

- Finalidades
- Características básicas
- Problemas de segurança